

Relation Algebra

§8. Thinning Algorithms

§8.1. Thinning

(8.1)

For $Q : A \rightarrow A$, $\text{thin}(Q) \triangleq (\exists/\exists) \cap (\epsilon \setminus (Q^\circ \epsilon)) : EA \rightarrow EA$ (8.1).

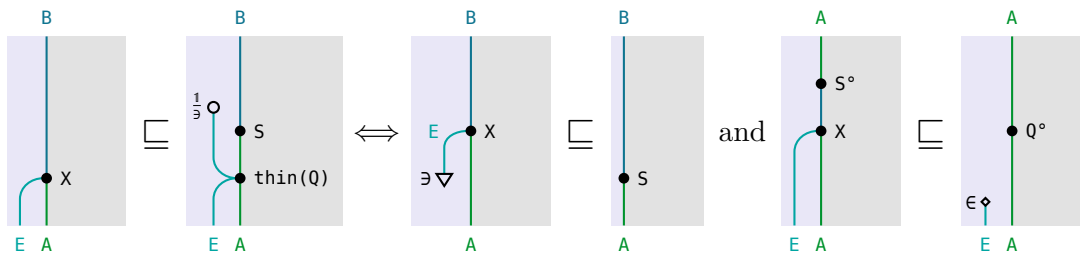
$ys \text{ thin}(Q) xs \iff xs \subseteq ys \wedge (\forall a \in ys. \exists b \in xs. b \ Q \ a)$ the same $^\circ$ as $\triangleq \text{est}$: ϵX runs $a \rightarrow ys \rightarrow xs$, member of ys first, and $Q^\circ \epsilon$ runs $a \rightarrow b \rightarrow xs$, so $(a, b) \in Q^\circ$ reads $b \ Q \ a$; at $Q \triangleq \leq$ every $a \in ys$ keeps some $b \leq a$ in xs , the end $\text{est}(\leq)$ picks

(8.1a)

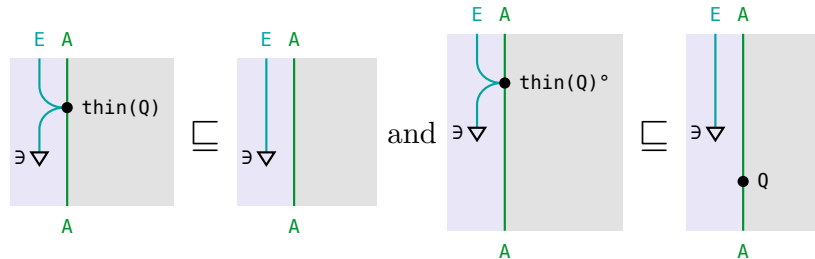
the law	what it says
$X \subseteq \frac{S}{\exists} \text{thin}(Q) \iff X \exists \subseteq S$ and $S^\circ X \subseteq Q^\circ \epsilon$	everything kept is an S -value, and every S -value has a Q -lower bound among the kept ones
$\text{thin}(Q) \exists \subseteq \exists$	everything a thinning keeps was in the set
$\text{thin}(Q)^\circ \exists \subseteq \exists Q$	every element of the set has a Q -lower bound among the kept ones
$Q \subseteq R \implies \text{thin}(Q) \subseteq \text{thin}(R)$	the fewer pairs Q relates, the fewer subsets count as thinnings
$1 \subseteq Q \implies 1 \subseteq \text{thin}(Q)$ $Q \subseteq Q \implies \text{thin}(Q) \text{thin}(Q) \subseteq \text{thin}(Q)$	keeping everything is always a legal thinning
$Q \subseteq R \wedge 1 \subseteq Q \wedge R R \subseteq R \implies \text{est}(R) = \text{thin}(Q) \text{est}(R)$ $Q \subseteq R, 1 \subseteq Q, R R \subseteq R$ — (8.1.2a); weaker than the book's “both preorders”: Q transitive is never used and $1 \subseteq R$ follows	thin-introduction : thinning first cannot lose an R -minimum
$\text{thin}(Q) \supseteq \text{est}(Q) \frac{1}{\exists}$ (8.2) — (8.2)	thin-elimination : keeping one element is a thinning, but its domain is the sets $\text{est}(Q)$ is defined on
$\frac{S}{\exists} \text{thin}(Q) \supseteq \frac{S}{\exists} \text{est}(R) \frac{1}{\exists}$ (8.3), $R \cap (S^\circ S) \subseteq Q$ — thin-elimination variant	the usable variant: R need only refine Q between values S gives one argument
$P(\text{thin}(Q)) \text{union} \subseteq \text{union thin}(Q)$ (8.4)	thinning each member set is a thinning of the union

§8.1.1. $X \subseteq (S^\circ \exists) \text{thin}(Q) \iff X \exists \subseteq S$ and $S^\circ X \subseteq Q^\circ \epsilon$

(8.1.1a)



(8.1.1b)



(8.1.1a) at $X \triangleq \text{thin}(Q)$, $S \triangleq \exists$: the left side is $\text{thin}(Q) \subseteq \text{thin}(Q)$, since $\exists \circ \exists = \wedge(\exists) = 1$

§8.1.2. $\text{est}(R) = \text{thin}(Q) \text{est}(R)$ given $Q \subseteq R, 1 \subseteq Q, R R \subseteq R$

$\text{est}(R) \sqsubseteq \text{thin}(Q) \text{ est}(R)$ (8.1.2a)

the $\sqsubseteq$ half: keeping everything is a thinning — $1 \sqsubseteq Q$

(a) $1 \sqsubseteq \text{thin}(Q)$ — (8.1a)

$\text{thin}(Q) \text{ est}(R) \sqsubseteq \exists$ (8.1.2b)

the $\sqsubseteq$ half, first condition of the UP of est at $X \triangleq \text{thin}(Q) \text{ est}(R)$ — est-up

(a) $\text{est}(R) \sqsubseteq \exists$ — est-laws

(b) $\text{thin}(Q) \sqsubseteq \exists$ — (8.1.1a)

$\in \text{thin}(Q) \text{ est}(R) \sqsubseteq R^\circ$ (8.1.2c)

the $\sqsubseteq$ half, second condition — $Q \sqsubseteq R$, R transitive

(a) $\in \text{thin}(Q) \sqsubseteq Q^\circ \in$ — (8.1.1a)

(b) $\in \text{est}(R) \sqsubseteq R^\circ$ — est-laws

(c) R transitive

$\text{est}(Q) \frac{1}{\exists} \sqsubseteq \text{thin}(Q)$ (8.2)

the singleton holding a Q -least member of a set is a thinning of that set

$\text{est}(Q) \frac{1}{\exists} \sqsubseteq \exists$

every member of the singleton is a member of the set

(a) $\triangleq \text{thin}, \exists/\exists$ half

(b) $\frac{1}{\exists} \sqsubseteq 1$ — pow-laws; $\text{est}(Q) \sqsubseteq \exists$ — est-laws

$\in \text{est}(Q) \frac{1}{\exists} \sqsubseteq Q^\circ \in$

the singleton's member stands in Q to every member of the set

(a) $\triangleq \text{thin}, \in \setminus (Q^\circ \in)$ half

(b) $\in \text{est}(Q) \sqsubseteq Q^\circ$ — est-up

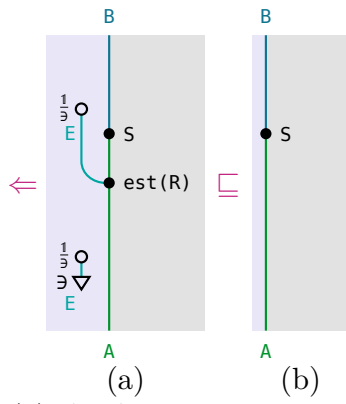
(c) $\frac{1}{\exists} \sqsubseteq \in$ — pow-laws

$$\text{Rn}S^\circ S \sqsubseteq Q \Rightarrow \frac{S}{\exists} \text{est}(R) \frac{1}{\exists} \sqsubseteq \frac{S}{\exists} \text{thin}(Q) \quad (8.3)$$

given $\text{Rn}(S^\circ S) \sqsubseteq Q$, Q a preorder

$$\frac{S}{\exists} \text{est}(R) \frac{1}{\exists} \exists \sqsubseteq S$$

every member of the singleton is a value S returns

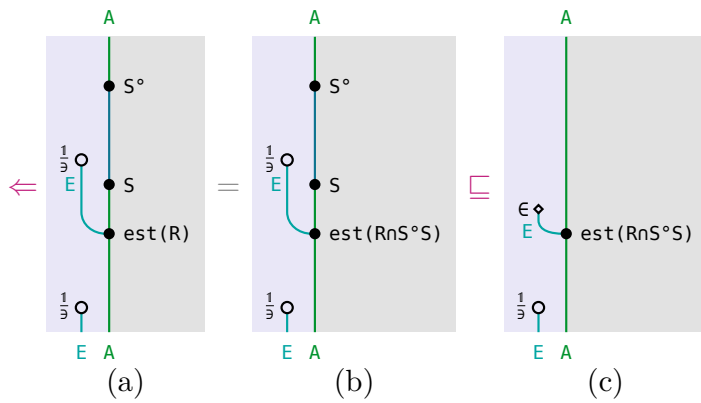


(a) (8.1a), first condition

(b) $\frac{1}{\exists} \exists = 1$ — pow-laws; $\frac{S}{\exists} \text{est}(R) \sqsubseteq S$ — est-laws

$$\text{Rn}S^\circ S \sqsubseteq Q \Rightarrow S^\circ \frac{S}{\exists} \text{est}(R) \frac{1}{\exists} \sqsubseteq Q^\circ \in$$

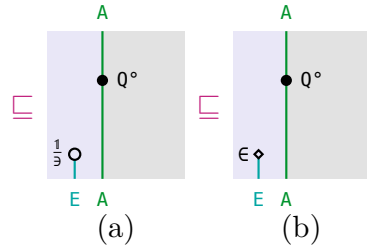
the singleton's member stands in Q to every value S returns at the same argument



(a) (8.1a), second condition

(b) $\frac{S}{\exists} \text{est}(R) = \frac{S}{\exists} \text{est}(RnS^\circ S)$ — est-laws

(c) $S^\circ \frac{S}{\exists} \sqsubseteq E$ — pow-laws



(a) $\in \text{est}(RnS^\circ S) \sqsubseteq (RnS^\circ S)^\circ$ — est-up; $\text{Rn}(S^\circ S) \sqsubseteq Q$

(b) $\frac{1}{\exists} \sqsubseteq E$ — pow-laws

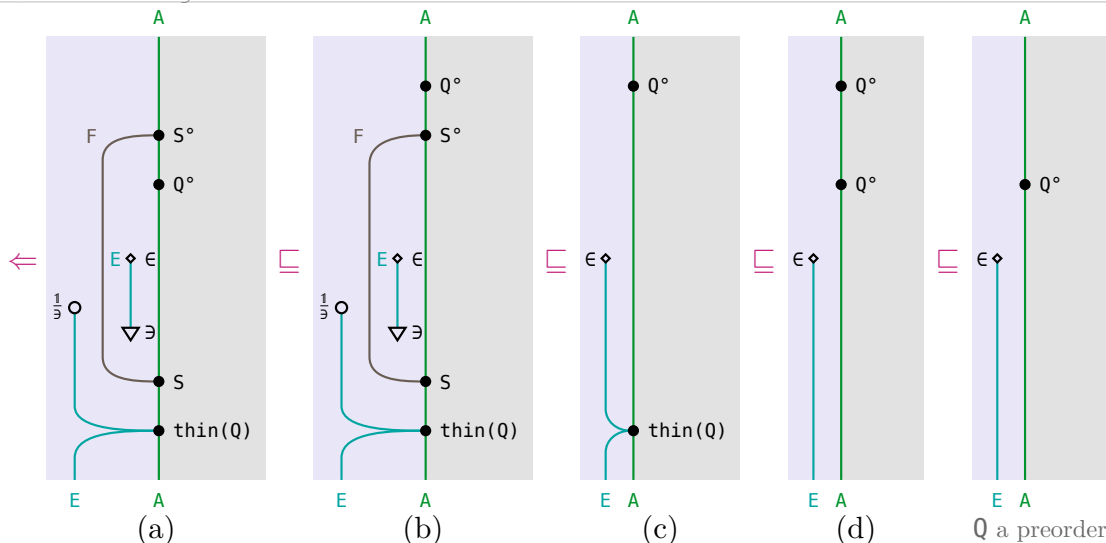
$$Q \sqsubseteq Q \wedge \text{MonoAlg}(S, Q) \Rightarrow \langle \frac{F(\exists)S}{\exists} \text{thin}(Q) \rangle \sqsubseteq \langle \frac{(S)}{\exists} \text{thin}(Q) \rangle$$

Theorem 8.1

thinning at every step of the reduce is a thinning of the whole candidate set — S monotonic on Q , Q a preorder

$$\langle \frac{F(\exists)S}{\exists} \text{thin}(Q) \rangle \sqsubseteq \langle \frac{(S)}{\exists} \text{thin}(Q) \rangle \text{ and } \langle \frac{(S)}{\exists} \rangle \langle \frac{F(\exists)S}{\exists} \text{thin}(Q) \rangle \sqsubseteq \langle \frac{(S)}{\exists} \rangle \langle \frac{(S)}{\exists} \text{thin}(Q) \rangle$$

(8.1a) at $X \triangleq \langle \frac{F(\exists)S}{\exists} \text{thin}(Q) \rangle$, $\langle (S) \rangle$ for its S

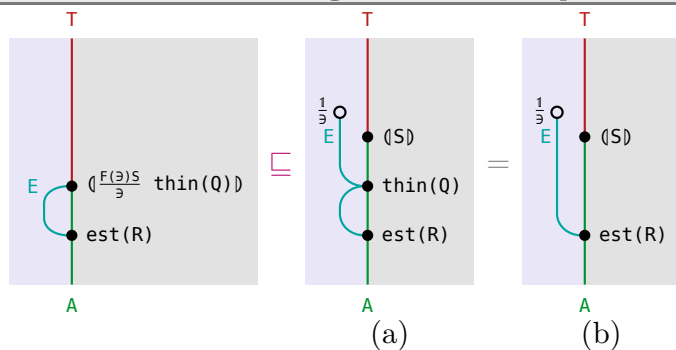


- (a) fusion, hylomorphism theorem
- (b) $S^\circ F(Q^\circ) \sqsubseteq Q^\circ S^\circ$ — mon-str, relator-laws
- (c) $S^\circ F(\epsilon) \frac{F(\exists)S}{\exists} \sqsubseteq \epsilon$ — pow-laws
- (d) $\epsilon \text{thin}(Q) \sqsubseteq Q^\circ \epsilon$ — $\triangleq \text{thin}$, adj-all

$$Q \sqsubseteq R \wedge 1 \sqsubseteq Q \wedge Q \sqsubseteq Q \wedge R \sqsubseteq R \wedge \text{MonoAlg}(S, Q) \Rightarrow \langle \frac{F(\exists)S}{\exists} \text{thin}(Q) \rangle \text{est}(R) \sqsubseteq \langle \frac{(S)}{\exists} \text{est}(R) \rangle$$

Corollary 8.1

the thinning fold refines the optimisation problem itself — S monotonic on Q , $Q \sqsubseteq R$, both preorders



- (a) thinning theorem
- (b) $\text{est}(R) = \text{thin}(Q) \text{est}(R)$ — (8.1a)

§8.2. Paths in a layered network

(8.2a)

$F(A, X) = A + A \times X$, $L = \text{list}^+$ with initial algebra $\alpha \triangleq [\text{wrap}, \text{cons}] : F(A, LA) \rightarrow LA$.

$\text{wrapz} \triangleq \langle \text{wrap}, \text{zero} \rangle$, $\text{consw}(a, (xs, n)) = (\text{cons}(a, xs), \text{wt}(a, \text{head}(xs)) + n)$.

$\text{cost} \triangleq \langle [\text{wrapz}, \text{consw}] \rangle \pi_2$, $\langle [\text{wrapz}, \text{consw}] \rangle = \langle 1, \text{cost} \rangle$, $R \triangleq \text{cost} \leq \text{cost}^\circ$.

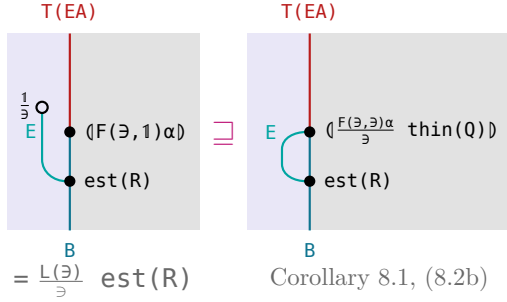
$Q \triangleq R \circ (\text{head} \text{head}^\circ)$, $S \triangleq F(1, \exists) \alpha$, $\frac{F(\exists, 1)}{\exists} = 1 + \text{cpl}$, $\frac{F(1, \exists)}{\exists} = 1 + \text{cpr}$, $\text{step} \triangleq \text{cpr } P(\text{cons}) \text{est}(R)$.

(8.2b)

the law	what it says
$F(\exists, Q) \alpha \sqsubseteq F(\exists, 1) \alpha Q$	$F(\exists, 1) \alpha$ is monotonic on Q ; on R it is not, since the next edge can cost arbitrarily much
$S \text{head} \sqsubseteq [1, \pi_1] \quad S \triangleq F(1, \exists) \alpha$	$S \text{head}$ is simple, which gives $R \circ (S^\circ S) \sqsubseteq Q$: between two paths S builds from one argument, equal cost and equal head already means Q

$$Q \sqsubseteq R \wedge 1 \sqsubseteq Q \wedge Q \sqsubseteq Q \wedge R \circ R \circ E \circ \wedge \text{MonoAlg}(F(\exists, 1)\alpha, Q) \wedge Rn(F(1, \exists)\alpha) \circ F(1, \exists)\alpha \sqsubseteq Q \Rightarrow \downarrow \frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists)\alpha}{\exists}) \text{est}(R) \text{est}(R) \sqsubseteq \frac{F(\exists, 1)\alpha}{\exists} \text{est}(R) \quad (8.2c)$$

a least-cost path in a layered network, as a fold over the layers



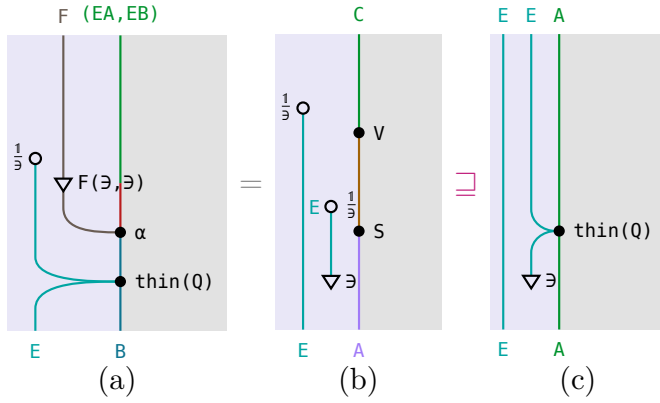
$\sqsubseteq$ (8.2d) under $\downarrow$ $\sqsupset$ monotonic: the whole chain runs inside the reduce, and the $\text{est}(R)$ behind it never moves.

$$Rn(F(1, \exists)\alpha) \circ F(1, \exists)\alpha \sqsubseteq Q \Rightarrow \frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists)\alpha}{\exists}) \text{est}(R) \sqsubseteq \frac{F(\exists, \exists)\alpha}{\exists} \text{thin}(Q) \quad (8.2c)$$

thinning the algebra of a layered network costs no more than taking the program's two cases

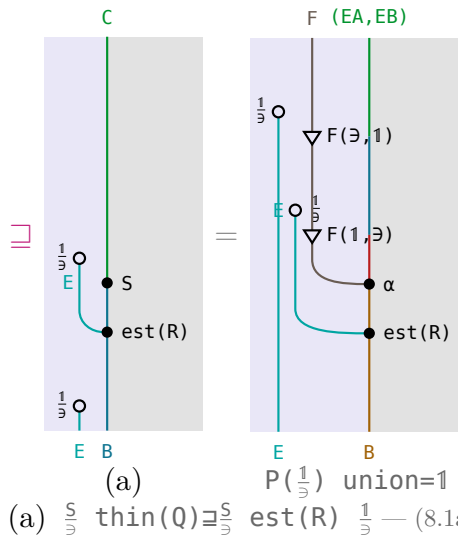
$$\frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists)\alpha}{\exists}) P(\text{thin}(Q)) \text{union} \sqsubseteq \frac{F(\exists, \exists)\alpha}{\exists} \text{thin}(Q)$$

thinning each set the algebra's parts build, then taking the union, is below thinning the one set the whole algebra builds



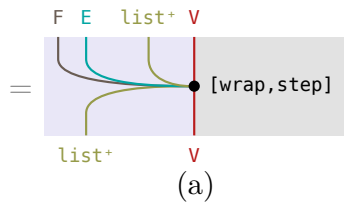
$$Rn(F(1, \exists)\alpha) \circ F(1, \exists)\alpha \sqsubseteq Q \Rightarrow \frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists)\alpha}{\exists}) \text{est}(R) \sqsubseteq \frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists)\alpha}{\exists}) P(\text{thin}(Q)) \text{union}$$

choosing an R-least member of each of those sets is below thinning each and taking the union



$$\text{cp } P(\alpha) \text{est}(\text{path}) = [\text{wrap}, \text{step}]$$

at the layered network, choosing an R-least path from each case is the program's two cases $[\text{wrap}, \text{step}]$



$$(a) \alpha \text{ a map; } \frac{F(1, \exists)}{\exists} P(\alpha) \text{est}(R) = [\text{wrap}, \text{step}] \text{---} \triangleq \text{cost}$$

$$= \frac{F(\exists, 1)}{\exists} = 1 + \text{cpl} \text{---} \triangleq \text{cost}$$

§8.3. Implementing thin

(8.3a)

$\text{setify} : [A] \rightarrow EA$, $\text{cup} : EA \times EA \rightarrow EA$, $\text{cp}(F) \triangleq \frac{F(\exists)}{\exists}$, $\text{listcp} : F(L) \rightarrow LF$, $\text{sort}(P) \triangleq \text{setify}^\circ \text{ ordered } P$ for P a connected preorder.

$\text{thinlist}(Q)$ is any $\text{thinlist}(Q) \sqsubseteq \text{subseq}$ with $\text{thinlist}(Q) \text{ setify} \sqsubseteq \text{setify thin}(Q)$; one is $\langle [\text{nil}, \text{bump } Q] \rangle$, $\text{bump } Q (a, []) = [a]$, $\text{bump } Q (a, [b] \# xs) = (b \text{ } Q \text{ } a \rightarrow [a] \# xs, a \text{ } Q \text{ } b \rightarrow [b] \# xs, [a] \# [b] \# xs)$.

Binary thinning data: $S = (f_1 p_1) \cup (f_2 p_2)$ with p_1, p_2 coreflexive; Q a preorder with $Q \sqsubseteq R$ and both $f_1 p_1, f_2 p_2$ monotonic on Q ; P a connected preorder with both f_1, f_2 monotonic on P ; $g_i \triangleq \text{list}(f_i) \text{ filter}(p_i)$.

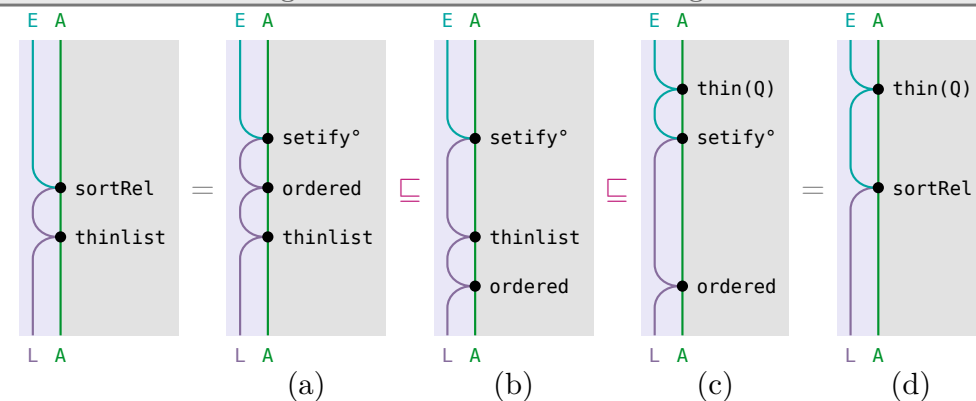
(8.3b)

the law	what it says
$\text{thinlist}(Q) \text{ xs} = [\text{minlist}(Q) \text{ xs}]$ (8.5), Q connected, xs non-empty	what thinning should come to when it can: one element
$\text{sort}(P) \text{ thinlist}(Q) \sqsubseteq \text{thin}(Q) \text{ sort}(P)$ (8.6) — (8.6)	thinning a sorted list is a thinning of the set — this is what $\text{thinlist}(Q) \sqsubseteq \text{subseq}$ buys
$\text{sort}(P) \text{ minlist}(Q) \sqsubseteq \text{est}(Q)$ (8.7)	a minimum of the sorted list is a minimum of the set
$\text{sort}(f P f^\circ) \text{ list}(f) \sqsubseteq P(f) \text{ sort}(P)$ (8.8)	shunt a function through a sort
$\text{sort}(P) \text{ filter}(p) \sqsubseteq E(p) \text{ sort}(P)$ (8.9), p coreflexive	filtering a sorted list sorts the restricted set
$(\text{sort}(P) \times \text{sort}(P)) \text{ merge}(P) \sqsubseteq \text{cup sort}(P)$ (8.10)	merging two sorted lists sorts their union
$F(\text{sort}(P)) \text{ listcp} \sqsubseteq \text{cp}(F) \text{ sort}(FP)$ (8.11), F linear	listcp is the list implementation of the cartesian product $\text{cp}(F)$

Map(setify) $\wedge$ Coreflexive(ordered) $\wedge$ thinlist $\sqsubseteq \text{subseq} \wedge$ ordered subseq $\sqsubseteq \text{subseq ordered} \wedge$ thinlist (8.6)

setify $\sqsubseteq \text{setify thin}(Q) \Rightarrow \text{sortRel thinlist} \sqsubseteq \text{thin}(Q) \text{ sortRel}$

a thinning of the sorted list lists a thinning of the set — P a connected preorder, $\text{thinlist}(Q) \sqsubseteq \text{subseq}$.



(a) $\text{sort}(P) \triangleq \text{setify}^\circ \text{ ordered } P \text{ — } \triangleq \text{thinlist}$

(b) $\text{ordered } P \text{ thinlist}(Q) \sqsubseteq \text{thinlist}(Q) \text{ ordered } P \text{ — } \triangleq \text{thinlist}$

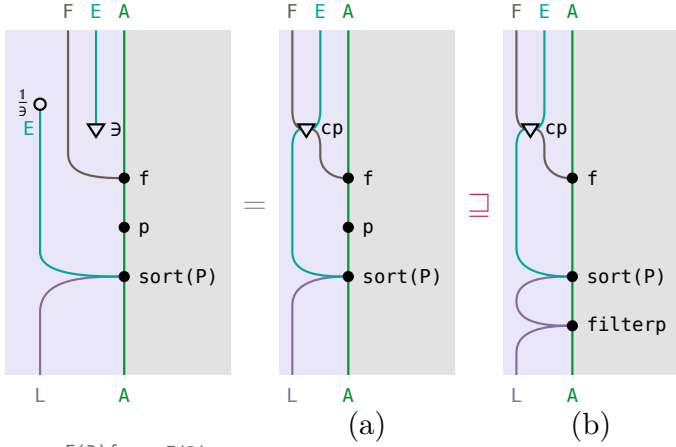
(c) $\text{thinlist}(Q) \text{ setify} \sqsubseteq \text{setify thin}(Q) \text{ — } \triangleq \text{thinlist}$, dom-laws, triple-chains

(d) $\text{sort}(P) \triangleq \text{setify}^\circ \text{ ordered } P \text{ — } \triangleq \text{thinlist}$

Lemma 8.1

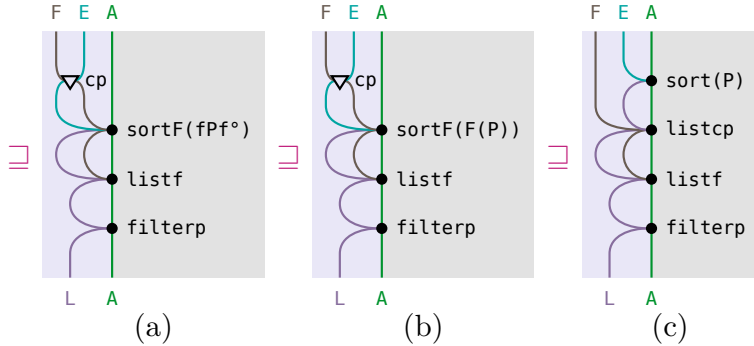
$\text{Map}(f) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{MonoAlg}(f, P) \wedge$
 $\text{sortF}(fP f^\circ) \text{listf} \sqsubseteq P(f) \text{sort}(P) \wedge \text{sort}(P) \text{filterp} \sqsubseteq E(p) \text{sort}(P) \wedge F(\text{sort}(P)) \text{listcp} \sqsubseteq \text{cp}$
 $\text{sortF}(F(P)) \Rightarrow F(\text{sort}(P)) \text{listcp} \text{listf} \text{filterp} \sqsubseteq \frac{F(\exists)fp}{\exists} \text{sort}(P)$
 one sorted list built from sorted arguments, instead of a set built and then sorted — $f : FA \rightarrow A$ monotonic
 on P , p coreflexive, F linear.

$\text{Map}(f) \wedge \text{sort}(P) \text{filterp} \sqsubseteq E(p) \text{sort}(P) \Rightarrow \text{cp } P(f) \text{sort}(P) \text{filterp} \sqsubseteq \frac{F(\exists)fp}{\exists} \text{sort}(P)$
 sorting the f -images of the product and filtering by p is below sorting the set $F(\exists)fp$ builds



- (a) $\frac{F(\exists)fp}{\exists} = \frac{F(\exists)}{\exists} E(fp)$ — pow-laws, $\triangleq$ thinlist
 $E(f) = P(f)$, f a map — powrel-laws
 (b) $\text{sort}(P) \text{filterp} \sqsubseteq E(p) \text{sort}(P)$ — (8.3b)

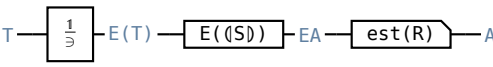
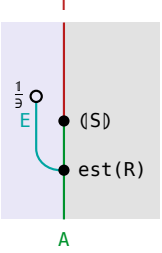
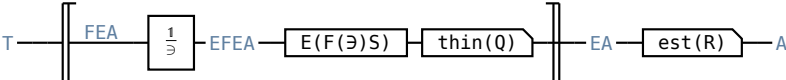
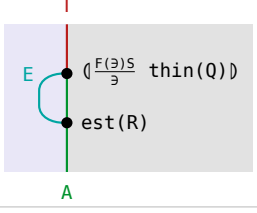
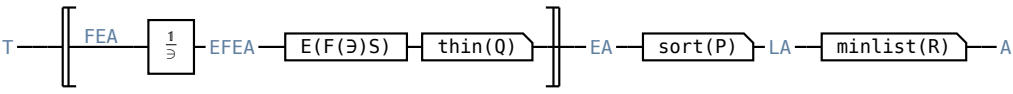
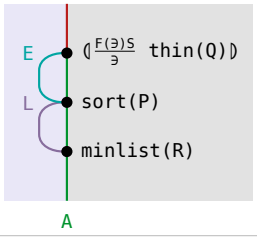
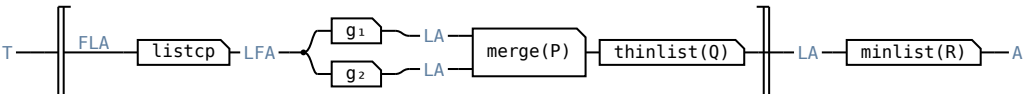
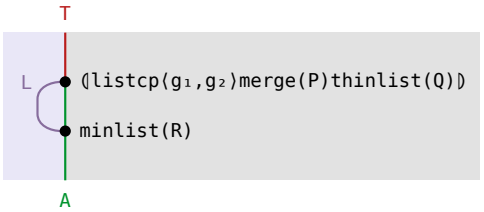
$\text{Map}(f) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{MonoAlg}(f, P) \wedge \text{sortF}(fP f^\circ) \text{listf} \sqsubseteq P(f) \text{sort}(P) \wedge$
 $F(\text{sort}(P)) \text{listcp} \sqsubseteq \text{cp} \text{sortF}(F(P)) \Rightarrow F(\text{sort}(P)) \text{listcp} \text{listf} \text{filterp} \sqsubseteq \text{cp } P(f) \text{sort}(P) \text{filterp}$
 combining the arguments' sorted lists by listcp , mapping f and filtering is below sorting the f -images and filtering



- (a) $\text{sort}(fP f^\circ) \text{list}(f) \sqsubseteq P(f) \text{sort}(P)$ — (8.3b)
 (b) $FP \sqsubseteq fP f^\circ$ — mon-str, $\triangleq$ thinlist
 (c) $F(\text{sort}(P)) \text{listcp} \sqsubseteq \text{cp}(F) \text{sort}(FP)$ — (8.3b)

$\text{Map}(f_1) \wedge \text{Map}(f_2) \wedge Q \subseteq R \wedge 1 \subseteq Q \wedge Q \subseteq Q \wedge R \circ R \subseteq R \wedge \text{MonoAlg}(f_1 \text{ } p_1, Q) \wedge \text{MonoAlg}(f_2 \text{ } p_2, Q)$ Theorem 8.2
 $\wedge (\forall X. \forall Y. X \subseteq Y \implies \text{sortF}(X) \subseteq \text{sortF}(Y)) \wedge \text{MonoAlg}(f_1, P) \wedge \text{MonoAlg}(f_2, P) \wedge \text{sortF}(f_1 \text{ } P f_1^\circ) \text{listf}_1 \subseteq P(f_1) \text{sort}(P) \wedge \text{sortF}(f_2 \text{ } P f_2^\circ) \text{listf}_2 \subseteq P(f_2) \text{sort}(P) \wedge$
 $\text{sort}(P) \text{filterp}_1 \subseteq E(p_1) \text{sort}(P) \wedge \text{sort}(P) \text{filterp}_2 \subseteq E(p_2) \text{sort}(P) \wedge F(\text{sort}(P)) \text{listcp} \subseteq \text{cp}$
 $\text{sortF}(F(P)) \wedge (\text{sort}(P) \times \text{sort}(P)) \text{merge}(P) \subseteq \text{cup } \text{sort}(P) \wedge$
 $\text{sort}(P) \text{thinlist}(Q) \subseteq \text{thin}(Q) \text{sort}(P) \wedge \text{sort}(P) \text{minlist}(R) \subseteq \text{est}(R) \wedge g_1 = \text{listf}_1 \text{ filterp}_1 \wedge$
 $g_2 = \text{listf}_2 \text{ filterp}_2 \wedge S = f_1 \text{ } p_1 \cup f_2 \text{ } p_2 \implies (\text{listcp}(g_1, g_2) \text{merge}(P) \text{thinlist}(Q))$
 $\text{minlist}(R) \subseteq \frac{(S)}{S} \text{est}(R)$

a fold on sorted lists of partial solutions, thinned at every step — at $\triangleq \text{thinlist}$'s binary thinning data.

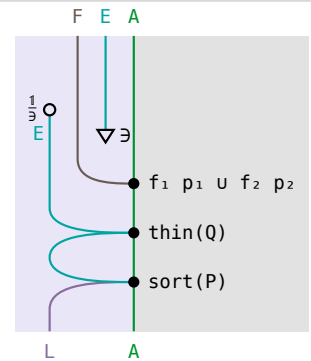
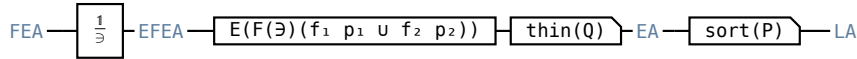
circuit	Hinze–Marsden
	
 Corollary 8.1 at $f_1 p_1$ and $f_2 p_2$ monotonic on $Q \triangleq \text{thinlist}$	
 $\text{sort}(P) \text{minlist}(R) \subseteq \text{est}(R) \text{ — (8.3b) at its } Q \triangleq R$	
 fusion at (8.3c)	

$$\begin{aligned}
& \text{Map}(f_1) \wedge \text{Map}(f_2) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{MonoAlg}(f_1, P) \wedge \text{MonoAlg}(f_2, P) \wedge \quad (8.3c) \\
& \text{sortF}(f_1 \text{ Pf}_1^\circ) \text{listf}_1 \sqsubseteq \text{P}(f_1) \text{sort}(P) \wedge \text{sortF}(f_2 \text{ Pf}_2^\circ) \text{listf}_2 \sqsubseteq \text{P}(f_2) \text{sort}(P) \wedge \\
& \text{sort}(P) \text{filterp}_1 \sqsubseteq \text{E}(p_1) \text{sort}(P) \wedge \text{sort}(P) \text{filterp}_2 \sqsubseteq \text{E}(p_2) \text{sort}(P) \wedge \text{F}(\text{sort}(P)) \text{listcp} \sqsubseteq \text{cp sortF}(\text{F}(P)) \\
& \wedge (\text{sort}(P) \times \text{sort}(P)) \text{merge}(P) \sqsubseteq \text{cup sort}(P) \wedge \text{sort}(P) \text{thinlist}(Q) \sqsubseteq \text{thin}(Q) \text{sort}(P) \Rightarrow \\
& \text{F}(\text{sort}(P)) \text{listcp}(\text{listf}_1 \text{ filterp}_1, \text{listf}_2 \text{ filterp}_2) \text{merge}(P) \text{thinlist}(Q) \sqsubseteq \frac{\text{F}(\exists)(f_1 \text{ p}_1 \cup f_2 \text{ p}_2)}{\exists} \\
& \text{thin}(Q) \text{sort}(P)
\end{aligned}$$

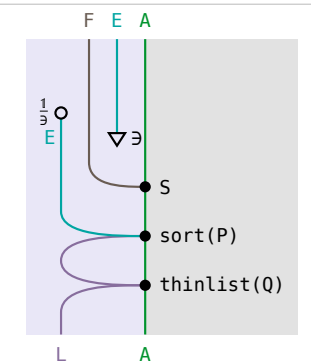
sorting the candidate set is what turns the thinning algebra into an algebra on lists — the side condition of binary thinning theorem's last step.

circuit

Hinze–Marsden



$$\begin{aligned}
& \text{FEA} \rightarrow \boxed{\frac{1}{3}} \rightarrow \text{EFEA} \rightarrow \boxed{\text{E}(\text{F}(\exists)S)} \rightarrow \text{EA} \rightarrow \text{sort}(P) \rightarrow \text{thinlist}(Q) \rightarrow \text{LA} \\
& \text{sort}(P) \text{ thinlist}(Q) \sqsubseteq \text{thin}(Q) \text{ sort}(P) \quad (8.3b)
\end{aligned}$$



$$\begin{aligned}
& \text{FEA} \rightarrow \boxed{\frac{1}{3}} \rightarrow \text{EFEA} \rightarrow \boxed{\text{E}(\text{F}(\exists)f_1 \text{ p}_1)} \rightarrow \text{EA} \rightarrow \text{cup} \rightarrow \text{EA} \rightarrow \text{sort}(P) \rightarrow \text{thinlist}(Q) \rightarrow \text{LA} \\
& = \left\langle \frac{\text{F}(\exists)f_1 \text{ p}_1}{\exists}, \frac{\text{F}(\exists)f_2 \text{ p}_2}{\exists} \right\rangle \text{cup sort}(P) \text{ thinlist}(Q) \\
& S = (f_1 \text{ p}_1) \cup (f_2 \text{ p}_2) \quad \triangleq \text{thinlist, then } \triangleq \text{cup}
\end{aligned}$$

$$\begin{aligned}
& \text{FEA} \rightarrow \boxed{\frac{1}{3}} \rightarrow \text{EFEA} \rightarrow \boxed{\text{E}(\text{F}(\exists)f_1 \text{ p}_1)} \rightarrow \text{EA} \rightarrow \text{sort}(P) \rightarrow \text{merge}(P) \rightarrow \text{thinlist}(Q) \rightarrow \text{LA} \\
& \text{FEA} \rightarrow \boxed{\frac{1}{3}} \rightarrow \text{EFEA} \rightarrow \boxed{\text{E}(\text{F}(\exists)f_2 \text{ p}_2)} \rightarrow \text{EA} \rightarrow \text{sort}(P) \rightarrow \text{merge}(P) \rightarrow \text{thinlist}(Q) \rightarrow \text{LA} \\
& = \left\langle \frac{\text{F}(\exists)f_1 \text{ p}_1}{\exists}, \frac{\text{F}(\exists)f_2 \text{ p}_2}{\exists} \right\rangle (\text{sort}(P) \times \text{sort}(P)) \text{ merge}(P) \text{ thinlist}(Q) \\
& (\text{sort}(P) \times \text{sort}(P)) \text{ merge}(P) \sqsubseteq \text{cup sort}(P) \quad (8.3b)
\end{aligned}$$

$$\begin{aligned}
& \text{FEA} \rightarrow \boxed{\frac{1}{3}} \rightarrow \text{EFEA} \rightarrow \boxed{\text{E}(\text{F}(\exists)f_1 \text{ p}_1)} \rightarrow \text{EA} \rightarrow \text{sort}(P) \rightarrow \text{merge}(P) \rightarrow \text{thinlist}(Q) \rightarrow \text{LA} \\
& \text{FEA} \rightarrow \boxed{\frac{1}{3}} \rightarrow \text{EFEA} \rightarrow \boxed{\text{E}(\text{F}(\exists)f_2 \text{ p}_2)} \rightarrow \text{EA} \rightarrow \text{sort}(P) \rightarrow \text{merge}(P) \rightarrow \text{thinlist}(Q) \rightarrow \text{LA} \\
& = \left\langle \frac{\text{F}(\exists)f_1 \text{ p}_1}{\exists} \text{ sort}(P), \frac{\text{F}(\exists)f_2 \text{ p}_2}{\exists} \text{ sort}(P) \right\rangle \text{ merge}(P) \text{ thinlist}(Q) \\
& \langle X, Y \rangle (\text{sort}(P) \times \text{sort}(P)) = \langle X \text{ sort}(P), Y \text{ sort}(P) \rangle \quad \text{— bdm-prod-laws}
\end{aligned}$$

$$\begin{aligned}
& \text{FEA} \rightarrow \boxed{\text{F}(\text{sort}(P))} \rightarrow \text{FLA} \rightarrow \text{listcp} \rightarrow \text{LFA} \rightarrow \left\langle \text{listf}_1, \text{filterp}_1 \right\rangle \rightarrow \text{LA} \rightarrow \text{merge}(P) \rightarrow \text{thinlist}(Q) \rightarrow \text{LA} \\
& \text{FEA} \rightarrow \boxed{\text{F}(\text{sort}(P))} \rightarrow \text{FLA} \rightarrow \text{listcp} \rightarrow \text{LFA} \rightarrow \left\langle \text{listf}_2, \text{filterp}_2 \right\rangle \rightarrow \text{LA} \rightarrow \text{merge}(P) \rightarrow \text{thinlist}(Q) \rightarrow \text{LA} \\
& \text{F}(\text{sort}(P)) \text{ listcp} \langle g_1, g_2 \rangle \text{ merge}(P) \text{ thinlist}(Q) \\
& \text{Lemma 8.1 at } f_1, p_1 \text{ and at } f_2, p_2, \text{ then } X \langle g_1, g_2 \rangle \sqsubseteq \langle Xg_1, Xg_2 \rangle \quad \text{— bdm-prod-laws; } g_i \triangleq \text{list}(f_i) \\
& \text{filter}(p_i) \quad \triangleq \text{thinlist}
\end{aligned}$$

§8.4. The knapsack problem

(8.4a)

`vol, wt : Item → Real, value ≐ list(vol) sum, weight ≐ list(wt) sum .`

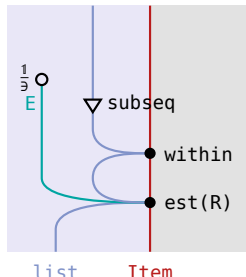
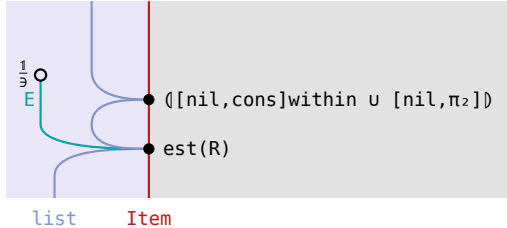
`subseq = ([nil, cons] ∪ [nil, π2]) ∘ , ,` within `w` the coreflexive on `xs` with weight `xs ≤ w`, `0 ≤ w`.

`R ≐ value ≥ value°`, `Q ≐ Rn(weight ≤ weight°)`, `P ≐ R`, .

`FA = 1 + Item × A`, `listcp = wrap + cpr`, `g1 ≐ list([nil, cons])` `filter(within w)` `= [list(nil), h1]`,
`g2 ≐ list([nil, π2]) = [list(nil), h2]`.

`h1 ≐ list(cons) filter(within w)`, `h2 ≐ list(π2)`.

$$\begin{aligned}
 & \theta \leq_w \wedge (\forall i. \theta \leq_{wt}(i)) \wedge (\forall X. \forall Y. X \leq Y \Rightarrow \text{sortF}(X) \leq \text{sortF}(Y)) \wedge \\
 & \text{sortF}([\text{nil}, \text{cons}]R[\text{nil}, \text{cons}]^\circ) \text{listf}_1 \leq_P ([\text{nil}, \text{cons}]) \text{sortP} \wedge \\
 & \text{sortF}([\text{nil}, \pi_2]R[\text{nil}, \pi_2]^\circ) \text{listf}_2 \leq_P ([\text{nil}, \pi_2]) \text{sortP} \wedge \text{sortP} \text{filterp}_1 \leq_E (\text{within}) \text{sortP} \wedge \\
 & F(\text{sortP}) \text{listcp} \leq_{cp} \text{sortF}(F(R)) \wedge (\text{sortP} \times \text{sortP}) \text{mergeP} \leq_{cup} \text{sortP} \wedge \text{sortP} \\
 & \text{thinlist} \leq_{\text{thin}} (Q(\text{vol}, \text{wt})) \text{sortP} \wedge \text{sortP} \text{minlist} \leq_{\text{est}} (R) \Rightarrow (\text{listcp}(\text{listf}_1 \text{filterp}_1, \text{listf}_2) \text{mergeP} \\
 & \text{thinlist}) \text{minlist} \leq_{\substack{\text{subseq} \\ \exists} \text{within}} \text{est}(R)
 \end{aligned}
 \tag{8.4c}$$

$[Item] \xrightarrow{\frac{1}{\exists}} E(\text{subseq within}) \xrightarrow{E[Item]} \text{est}(R) \xrightarrow{[Item]}$ $\frac{\text{subseq (within w)}}{\exists} \text{est}(R)$	
$[Item] \xrightarrow{\frac{1}{\exists}} E(\langle [nil, cons] \text{ within } u \ [nil, \pi_2] \rangle) \xrightarrow{E[Item]} \text{est}(R) \xrightarrow{[Item]}$ $= \frac{\langle [nil, cons] \text{ (within w)} \ u \ [nil, \pi_2] \rangle}{\exists} \text{est}(R)$ fusion, weights non-negative.	
$[Item] \xrightarrow{\left[\begin{array}{c} \text{Item} \\ \text{list}[Item] \end{array} \right]} \text{listcp} \xrightarrow{\text{list}(F[Item])} \left[\begin{array}{c} \text{listf}_1 \quad \text{filterp}_1 \\ \text{listf}_2 \end{array} \right] \xrightarrow{\text{list}[Item]} \text{mergeP} \xrightarrow{\text{thinlist}} \left[\begin{array}{c} \text{list}[Item] \\ \text{minlist} \end{array} \right] \xrightarrow{[Item]}$ $\sqsubseteq \langle \text{listcp} \langle g_1, g_2 \rangle \text{ merge } R \text{ thinlist}(Q) \rangle \text{ minlist}(R)$ binary thinning theorem, at $P \triangleq R$, F linear, Q from (8.4b)	
$\langle [nil, cpr \langle h_1, h_2 \rangle \text{ merge } R \text{ thinlist}(Q)] \rangle \text{ minlist}(R)$ $= \text{listcp} = \text{wrap} + \text{cpr}, g_i = [\text{list}(nil), h_i] \xrightarrow{\triangle} \text{within}; \text{minlist}(R)$ is head, packings coming out in descending value	

§8.5. The paragraph problem

(8.5a)

$\text{Line} = \text{list}^+ \text{ Word}$, $\text{Para} = \text{list}^+ \text{ Line}$, $\text{FA} = \text{Word} + \text{Word} \times \text{A}$, $\text{listcp} = \text{wrap} + \text{cpr}$.
 $\text{new}(a, xs) = [[a]] \# xs$, $\text{glue}(a, xs) = [[a] \# \text{head}(xs)] \# \text{tail}(xs)$, $\text{partition} \triangleq ([\text{wrap} \ \text{wrap}, \text{new} \ \cup \ \text{glue}]) : \text{list}^+ \text{ Word} \rightarrow \text{Para}$.
 $\text{width} \triangleq ([\text{length}, (\text{length} \times 1) \text{ plus succ}])$, $0 \leq \text{length } a$, $\text{fits } w$ the coreflexive on a line x with width $x \leq w$, $\text{ok } w$ the coreflexive on $[x] \# xs$ with width $x \leq w$.
 $\text{white } w \ x = w - \text{width } x$, $\text{collect} \triangleq \text{list}(\text{sqr}) \text{ sum}$, $\text{waste } w \triangleq \text{init list}(\text{white } w) \text{ collect}$.
 $R \triangleq (\text{waste } w) \leq (\text{waste } w)^\circ$, $Q \triangleq R \cap (\text{head head}^\circ)$, $P \triangleq \tau$, .
 $g_1 \triangleq \text{list}([\text{wrap} \ \text{wrap}, \text{new}])$, $g_2 \triangleq \text{list}([\text{wrap} \ \text{wrap}, \text{glue}]) \text{ filter}(\text{ok } w)$, $\text{start} \triangleq \text{wrap} \ \text{wrap} \ \text{wrap}$, $h_1 \triangleq \text{list}(\text{new})$, $h_2 \triangleq \text{list}(\text{glue}) \text{ filter}(\text{ok } w)$.

(8.5b)

the law	what it says
$(1 \times R) \text{ glue} \sqsubseteq \text{glue } R \quad \text{FALSE}$	glue is not monotonic on R: the waste of a paragraph depends on its whole first line, so no greedy algorithm solves this
$(1 \times Q) \text{ new} \sqsubseteq \text{new } Q$ $(1 \times Q) (\text{glue } (\text{ok } w)) \sqsubseteq \text{glue } (\text{ok } w) Q$ cons monotonic on $\text{collect} \leq \text{collect}^\circ$,	both halves are monotonic on Q once ties in waste are broken by the first line
$\text{merge } \tau = \text{cat}$; $P \triangleq \text{head prefix head}^\circ$ also serves	τ needs no sorting at all, and prefix is a linear order on first lines of paragraphs of one input

(8.5c)

$(\forall a. 0 \leq \text{len}(a)) \wedge (\forall a. \text{len}(a) \leq w) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{sortF}([\text{wrap} \ \text{wrap}, \text{new}] \tau [\text{wrap} \ \text{wrap}, \text{new}]^\circ) \text{listf}_1 \sqsubseteq P([\text{wrap} \ \text{wrap}, \text{new}]) \text{sortP} \wedge \text{sortF}([\text{wrap} \ \text{wrap}, \text{glue}] \tau [\text{wrap} \ \text{wrap}, \text{glue}]^\circ) \text{listf}_2 \sqsubseteq P([\text{wrap} \ \text{wrap}, \text{glue}]) \text{sortP} \wedge \text{sortP filterp}_2 \sqsubseteq E(\text{ok}) \text{sortP} \wedge F(\text{sortP}) \text{listcp} \sqsubseteq \text{cp} \text{sortF}(F(\tau)) \wedge (\text{sortP} \times \text{sortP}) \text{mergeP} \sqsubseteq \text{cup sortP} \wedge \text{sortP thinlist} \sqsubseteq \text{thin}(Q(\text{len}, w)) \text{sortP} \wedge \text{sortP minlist} \sqsubseteq \text{est}(R) \Rightarrow ([\text{listcp}(\text{listf}_1, \text{listf}_2 \text{ filterp}_2) \text{ mergeP thinlist}] \text{minlist} \sqsubseteq \frac{\text{partition fits}}{3} \text{est}(R))$
 a paragraph laid out as a fold that thins the layouts kept at each word

circuit	Hinze–Marsden
$\text{list}^+(\text{Word}) \xrightarrow{\frac{1}{3}} E(\text{list}^+(\text{Word})) \xrightarrow{E(\text{partition fits})} E(\text{list}^+(\text{list}^+(\text{Word}))) \xrightarrow{\text{est}(R)} \text{list}^+(\text{list}^+(\text{Word}))$ $\frac{\text{partition list}^+(\text{fits } w)}{3} \text{est}(R)$	
$\text{list}^+(\text{Word}) \xrightarrow{\frac{1}{3}} E(\text{list}^+(\text{Word})) \xrightarrow{E([\text{wrap} \ \text{wrap}, \text{new}] \cup [\text{wrap} \ \text{wrap}, \text{glue}] \text{ok})} E(\text{list}^+(\text{list}^+(\text{Word}))) \xrightarrow{\text{est}(R)} \text{list}^+(\text{list}^+(\text{Word}))$ $= \frac{([\text{wrap} \ \text{wrap}, \text{new}] \cup (\text{glue } (\text{ok } w)))}{3} \text{est}(R)$ fusion, every word fits on a line by itself.	
$\text{list}^+(\text{Word}) \xrightarrow{\frac{1}{3}} E(\text{list}^+(\text{Word})) \xrightarrow{E([\text{wrap} \ \text{wrap}, \text{new}] \cup [\text{wrap} \ \text{wrap}, \text{glue}] \text{ok})} E(\text{list}^+(\text{list}^+(\text{Word}))) \xrightarrow{\text{est}(R)} \text{list}^+(\text{list}^+(\text{Word}))$ $= \frac{([\text{wrap} \ \text{wrap}, \text{new}] \cup ([\text{wrap} \ \text{wrap}, \text{glue}] (\text{ok } w)))}{3} \text{est}(R)$ the algebra as $(f_1 p_1) \cup (f_2 p_2)$, $p_1 \triangleq 1 \text{ —} \triangleq \text{thinlist}$.	
$\sqsubseteq ([\text{listcp } (g_1, g_2) \text{ cat thinlist}(Q)] \text{minlist}(R))$ binary thinning theorem, at $P \triangleq \tau$ with merge $\tau = \text{cat}$, Q from (8.5b)	
$Q[\text{start}, \text{cpr } (h_1, h_2) \text{ cat thinlist}(Q)]$ $= \text{minlist}(R)$ $\text{listcp} = \text{wrap} + \text{cpr}$, g_i along the coproduct — $\triangleq \text{partition}$	

